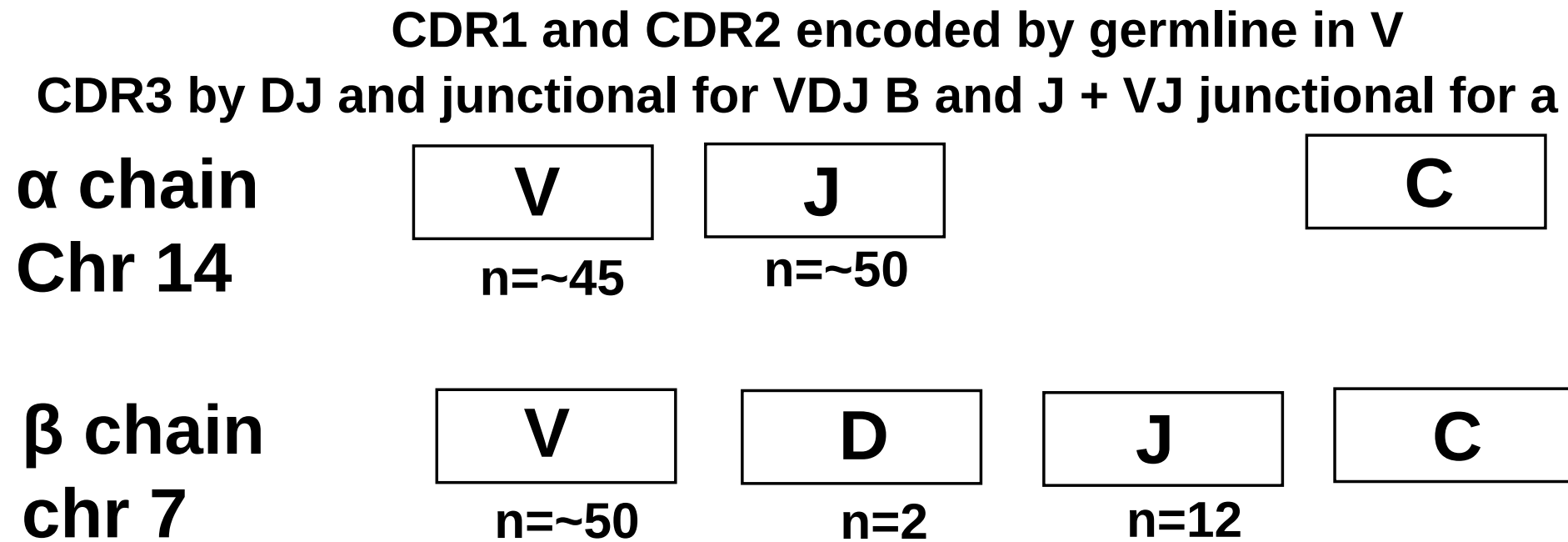




junctional diversity
(non-germline)

largest contributor to diversity

P nucleotides (template after hairpin opening by Artemis)

N nucleotides (then nontemplate followed by DNA ligation) (20 by TdT) - more common in B chain

combinatorial diversity
(germline)

1 to 3 million upper limit)

10^{15} - 10^{20} unique TCR combinations